

Board Name Fabrication Document

Layer Stack Legend

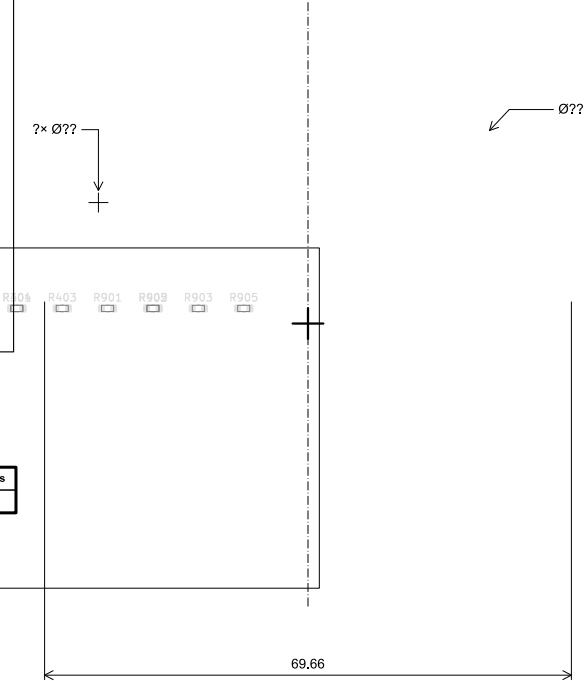
Material	Layer	Thickness	Dielectric	Type	Gerber
F,Paste				Paste Mask	
F,Silkscreen				Legend	GBR
F,Mask		0,02mm	Solder Resist	Solder Mask	GBR
Copper	L1 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
Prepreg		0,18mm	FR4_7628	Dielectric	
Copper	L2 (GND)	0,035mm (1,00oz)		Plane	GBR
Core		0,4mm	FR4	Dielectric	
Copper	L3 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Prepreg		0,18mm	FR4_7628	Dielectric	
Copper	L4 (Sig, PWR)	0,035mm (1,00oz)		Signal	GBR
Core		0,4mm	FR4	Dielectric	
Copper	L5 (GND)	0,035mm (1,00oz)		Plane	GBR
Prepreg		0,18mm	FR4_7628	Dielectric	
Copper	L6 (Sig, PWR)	0,07mm (2,00oz)		Signal	GBR
B,Mask		0,02mm	Solder Resist	Solder Mask	GBR
B,Silkscreen			Direct Printing	Legend	GBR
B,Paste				Paste Mask	

Total thickness: 1,66mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
Edge-Coupled Coated Microstrip	100	±10 %	L1	0,2032	0,28	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

- 1) FABRICATE PER IPC-6012A CLASS 2.
- 2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

DIMENSIONS OF CIRCUMSCRIBED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.
- 4) SURFACE FINISH: IMMERSION GOLD
- 5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR BLACK.
- 6) SILKSCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING YELLOW NON-CONDUCTIVE EPOXY INK.
- 7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK IS OPENED IN THE GERBER FILES.
- 8) VENDOR SHOULD FOLLOW RoHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING
- 9) PCB MATERIAL REQUIREMENTS:

A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.

B. Tg 170 C OR EQUIVALENT.

C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY ETH ROBOTICS CLUB.
- 10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE70,000 × 45,000 mm

BOARD THICKNESS1,660 mm

TRACE WIDTHN/A mm

TRACE TO TRACE0,200 mm

MIN. HOLE (PTH)0,300 mm

MIN. HOLE (NPTH)N/A mm

ANNULAR RING0,150 mm

COPPER TO HOLE0,254 mm

COPPER TO EDGE0,250 mm

HOLE TO HOLE0,254 mm
- 11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.
- 12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

	Comments:	Company: ETH Robotics Club		Variant: RELEASED	Git Hash: 1d53ab9
		Board Name: Board Name		Project Name: Project Name	
	Sheet Title: Top Fabrication (Scale 1:1)	File Name: ci_dummy.kicad_pcb	Designer: Author	Date: 2024-04-13	Revision: 0.2.0
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